

DATABASE SYSTEMS

CHAP. 3: Database design concepts and Normalization Theory

Database Normalization

- Database normalization is a database schema design technique, by which an existing schema is modified to minimize redundancy and dependency of data.

Benefits of Normalization

- Less storage space
- Quicker updates
- Less data inconsistency
- Clearer data relationships
- Easier to add data
- Flexible Structure

Functional Dependency

- Relationship that exists when one attribute uniquely determines another attribute.
- If r is a relation with attributes x and y , a functional dependency between the attributes is represented as $x \rightarrow y$, which specifies y is functionally dependent on x .
- The attributes of a table is said to be dependent on each other when an attribute of a table uniquely identifies another attribute of the same table.

Functional Dependency

- Describes relationships between attributes
- (Columns) in a relation (table) and is a property of the semantics of attributes
- An attribute A is functionally dependent on a set of attributes X if and only if
 - Value of A is determined solely by the values of X
 - Values of X uniquely determine a value of A
- $X \rightarrow a$
- $\text{Child} \rightarrow \text{mother}$
- $\text{Mother} \rightarrow \text{child}$

Types of Functional Dependencies

- Trivial functional dependency
- Non-trivial functional dependency
- Multivalued dependency
- Transitive dependency

Trivial functional dependency

- If the set of attributes includes that attribute.
- Symbolically:
 - $A \rightarrow B$ is trivial functional dependency if B is a subset of A .
 - The following dependencies are also trivial: $A \rightarrow A$ & $B \rightarrow B$

Trivial functional dependency

- example:
- Consider a table with two columns Student_id and Student_Name.
- $\{Student_Id, Student_Name\} \rightarrow Student_Id$ is a trivial functional dependency as Student_Id is a subset of $\{Student_Id, Student_Name\}$.
- Also, $Student_Id \rightarrow Student_Id$ & $Student_Name \rightarrow Student_Name$ are trivial dependencies too.

Non-trivial functional dependency

- If a functional dependency $X \rightarrow Y$ holds true where Y is not a subset of X then this dependency is called non trivial Functional dependency.

Non-trivial functional dependency

- example:
- An employee table with three attributes: emp_id, emp_name, emp_address.
- The following functional dependencies are non-trivial:
 - emp_id $\rightarrow$ emp_name
 - emp_name is not a subset of emp_id
 - emp_id $\rightarrow$ emp_address
 - emp_address is not a subset of emp_id

Multivalued dependency

- Multivalued dependency occurs when there are more than one independent multivalued attributes in a table.
- multivalued dependency is a full constraint between two sets of attributes in a relation.
- In contrast to the functional dependency, the multivalued dependency requires that certain tuples be present in a relation.

Multivalued dependency

- Consider a bike manufacture company, which produces two colors (Black and white) in each model every year.
- Here columns `manuf_year` and `color` are independent of each other and dependent on `bike_model`.
- In this case these two columns are said to be multivalued dependent on `bike_model`.
- These dependencies can be represented like this:
- `bike_model ->> manuf_year`
- `bike_model ->> color`

Transitive dependency

- if it is indirectly formed by two functional dependencies.
- $X \rightarrow Z$ is a transitive dependency if the following three functional dependencies hold true: $X \rightarrow Y$ Y does not $\rightarrow X$ $Y \rightarrow Z$
- A transitive dependency can only occur in a relation of three or more attributes.
- This dependency helps us normalizing the database in 3NF (3rd Normal Form).

Transitive dependency

- Example :
- $\{\text{Book}\} \rightarrow \{\text{Author}\}$
 - if we know the book, we know the author name
- $\{\text{Author}\}$ does not $\rightarrow \{\text{Book}\}$
- $\{\text{Author}\} \rightarrow \{\text{Author_age}\}$
- Therefore as per the rule of transitive dependency:
- $\{\text{Book}\} \rightarrow \{\text{Author_age}\}$ should hold, that makes sense because if we know the book name we can know the author's age.

Flat table

<u>EmpNo</u>	Name	<u>DoB</u>	<u>DeptID</u>	<u>DeptName</u>	Manager
3456	McTavish	14/9/1971	3	Personnel	Morgan
9012	Scott	5/3/1979	1	Sales	Jones
5678	Campbell	10/2/1976	2	Finance	Evans
1234	McDonald	2/3/1981	1	Sales	Jones

Flat table

Employee (EmpNo, Name, DoB, DeptID, DeptName, Manager),

The primary key is EmpNo. It has functional dependencies

- EmpNo → Name, DoB, DeptID, DeptName, Manager
- DeptID → DeptName, Manager

Note: DeptName and Manager are transitive dependencies on EmpNo

Flat table

Employee (EmpNo, Name, DoB, DeptID, DeptName, Manager),

The primary key is EmpNo. It has functional dependencies

- EmpNo → Name, DoB, DeptID, DeptName, Manager
- DeptID → DeptName, Manager

Note: DeptName and Manager are transitive dependencies on EmpNo

Problems with the flat table

- **Data redundancy or duplication**
- **Consequences of redundancy**
 - Wasted space
 - Potential performance cost
 - Potential inconsistency
 - Inability to represent data

Anomalies

- *Insertion anomalies*

- If a new employee is added to deptid 2, all the info about the department (name & manager) has to be re-input
- A new department, say IT, can be added only when there exists an employee for the new department

Anomalies

- ***Deletion anomalies***

- If the last employee from a department is deleted, then details of that department disappear from the database

- ***Update anomalies***

- If the manager of sales, jones, is replaced by smith, all the records for employees in sales department must be modified.

Flat table

Patient #	Surgeon #	Surg. date	Patient Name	Patient Addr	Surgeon	Surgery	Postop drug	Drug side effects
1111	145 311	Jan 1, 1995; June 12, 1995	John White	15 New St. New York, NY	Beth Little Michael Diamond	Gallstones <u>removal</u> ; Kidney stones removal	Penicillin, none-	Rash none
1234	243 467	Apr 5, 1994 May 10, 1995	Mary Jones	10 Main St. Rye, NY	Charles Field Patricia Gold	Eye Cataract removal Thrombosis removal	Tetracycline none	Fever none
2345	189	Jan 8, 1996	Charles Brown	Dogwood Lane Harrison, NY	David Rosen	Open Heart Surgery	Cephalosporin	none
4876	145	Nov 5, 1995	Hal Kane	55 Boston Post Road, Chester, CN	Beth Little	Cholecystectomy	<u>Demicillin</u>	none
5123	145	May 10, 1995	Paul Kosher	Blind Brook Mamaroneck, NY	Beth Little	Gallstones Removal	none	none
6845	243	Apr 5, 1994 Dec 15, 1984	Ann Hood	Hilton Road Larchmont, NY	Charles Field	Eye Cornea Replacement Eye cataract removal	Tetracycline	Fever

First normal form -1NF

- A relation is said to be in first normal form (1NF) if all attribute values are atomic: no repeating group, no composite attributes. Primary key is identified.

First normal form -1NF

Patient #	Surgeon #	Surgery Date	Patient Name	Patient Addr	Surgeon Name	Surgery	Drug admin	Side Effects
1111	145	01-Jan-95	John White	15 New St. New York, NY	Beth Little	Gallstones removal	Penicillin	rash
1111	311	12-Jun-95	John White	15 New St. New York, NY	Michael Diamond	Kidney stones removal	none	none
1234	243	05-Apr-94	Mary Jones	10 Main St. Rye, NY	Charles Field	Eye Cataract removal	Tetracycline	Fever
1234	467	10-May-95	Mary Jones	10 Main St. Rye, NY	Patricia Gold	Thrombosis removal	none	none
2345	189	08-Jan-96	Charles Brown	Dogwood Lane Harrison, NY	David Rosen	Open Heart Surgery	Cephalosporin	none
4876	145	05-Nov-95	Hal Kane	55 Boston Post Road, Chester, CN	Beth Little	Cholecystectomy	Demicillin	none
5123	145	10-May-95	Paul Kosher	Blind Brook Mamaroneck, NY	Beth Little	Gallstones Removal	none	none
6845	243	05-Apr-94	Ann Hood	Hilton Road Larchmont, NY	Charles Field	Eye Cornea Replacement	Tetracycline	Fever
6845	243	15-Dec-84	Ann Hood	Hilton Road Larchmont, NY	Charles Field	Eye cataract removal	none	none

Second normal form -2NF

- A relation R is in 2NF if
 - R is 1NF , and
 - All non-prime attributes are fully dependent on the candidate keys.Which is creating relationships between these new tables and their predecessors through the use of foreign keys.

Second normal form -2NF

- Partial dependency must be eliminated
- Break the composite primary key into two parts, each part representing a separate table
 - A prime attribute appears in a candidate key.
 - There is no partial dependency in 2NF.

Second normal form -2NF

Patient #	Surgeon #	Surgery Date	Surgery	Drug Admin	Side Effects
1111	145	01-Jan-95	Gallstones removal	Penicillin	rash
1111	311	12-Jun-95	Kidney stones removal	none	none
1234	243	05-Apr-94	Eye Cataract removal	Tetracycline	Fever
1234	467	10-May-95	Thrombosis removal	none	none
2345	189	08-Jan-96	Open Heart Surgery	Cephalosporin	none
4876	145	05-Nov-95	Cholecystectomy	<u>Demicillin</u>	none
5123	145	10-May-95	Gallstones Removal	none	none
6845	243	15-Dec-84	Eye cataract removal	none	none
6845	243	05-Apr-94	Eye Cornea Replacement	Tetracycline	Fever

Second normal form -2NF

Patient #	Patient Name	Patient Address
1111	John White	15 New St. New York, NY
1234	Mary Jones	10 Main St. Rye, NY
2345	Charles Brown	Dogwood Lane Harrison, NY
4876	Hal Kane	55 Boston Post Road, Chester, CN
5123	Paul Kosher	Blind Brook Mamaroneck, NY
6845	Ann Hood	Hilton Road Larchmont, NY

Surgeon #	Surgeon Name
145	Beth Little
189	David Rosen
243	Charles Field
311	Michael Diamond
467	Patricia Gold

Third normal form -3NF

- A relation is said to be in Third Normal Form if there is no transitive functional dependency between non-key attributes

Third normal form -3NF

Patient #	Surgeon #	Surgery Date	Surgery	Drug Admin
1111	145	01-Jan-95	Gallstones removal	Penicillin
1111	311	12-Jun-95	Kidney stones removal	none
1234	243	05-Apr-94	Eye Cataract removal	Tetracycline
1234	467	10-May-95	Thrombosis removal	none
2345	189	08-Jan-96	Open Heart Surgery	Cephalosporin
4876	145	05-Nov-95	Cholecystectomy	<u>Demicillin</u>
5123	145	10-May-95	Gallstones Removal	none
6845	243	15-Dec-84	Eye cataract removal	none
6845	243	05-Apr-94	Eye Cornea Replacement	Tetracycline

Third normal form -3NF

Drug Admin	Side Effects
Cephalosporin	none
<u>Demicillin</u>	none
none	none
Penicillin	rash
Tetracycline	Fever

Boyce-Codd form -3NF

- A 3NF relation is NOT in BCNF if:
 - Candidate keys in the relation are composite keys (they are not single attributes)
 - There is more than one candidate key in the relation, and
 - The keys are not disjoint, that is, some attributes in the keys are common

Boyce-Codd form -3NF

Patient #	Patient Name
1111	John White
1234	Mary Jones
2345	Charles Brown
4876	Hal Kane
5123	Paul Kosher
6845	Ann Hood

Patient #	Patient Address
1111	15 New St. New York, NY
1234	10 Main St. Rye, NY
2345	Dogwood Lane Harrison, NY
4876	55 Boston Post Road, Chester, CN
5123	Blind Brook Mamaroneck, NY
6845	Hilton Road Larchmont, NY

Fourth form -4NF

- Any relation is in Fourth Normal Form if it is BCNF *and* any multivalued dependencies are trivial.
- Eliminate non-trivial multivalued dependencies by projecting into simpler tables.

Fifth form -5NF

- A relation is in 5NF if every join dependency in the relation is implied by the keys of the relation
- Implies that relations that have been decomposed in previous NF can be recombined via natural joins to recreate the original relation.